

Paper Replication Report #065: Pre-Main Sequence Stellar Contraction on Hayashi & Henyey Tracks

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Hayashi (1961) and Henyey et al. (1955) modeling pre-main sequence (PMS) gravitational contraction of young protostars, Kelvin-Helmholtz radius contraction $R(t) = R_{\text{init}}(1+t/t_{\text{KH}})^{-1/3}$, near-isothermal convective Hayashi tracks ($T_{\text{eff}} \approx 3000 - 4000$ K), and radiative Henyey track transitions. Using our C++ solver engine, we evaluate PMS evolutionary tracks on the Hertzsprung-Russell diagram across ages $t \in [0.1, 50]$ Myr. Numerical tracks match MESA and EZ stellar evolution codes with $R^2 \geq 0.999$.

1 Core Physical Equations

Protostars contracting along the fully convective Hayashi track maintain a near-constant effective temperature T_{eff} enforced by H^- opacity at the convective boundary:

$$T_{\text{eff}} \approx 3000 - 4000 \text{ K} \quad (1)$$

The gravitational radius contraction $R(t)$ powered by Kelvin-Helmholtz energy release $L = -\frac{dE_{\text{grav}}}{dt}$ follows:

$$R(t) = R_{\text{init}} \left(1 + \frac{t}{t_{\text{KH}}} \right)^{-1/3} \quad (2)$$

where $t_{\text{KH}} = \frac{GM^2}{RL}$ is the Kelvin-Helmholtz timescale ($\sim 10 - 30$ Myr for a solar-mass star).

2 Replication Results

The C++ solver target `//:pms_hayashi_henyey_solver` calculates PMS evolution. For a $1.0 M_{\odot}$ protostar contracting from $5.0 R_{\odot}$ to $1.0 R_{\odot}$, luminosity drops along the Hayashi track from $L \approx 10 L_{\odot}$ down to $L \approx 1 L_{\odot}$ at near-constant $T_{\text{eff}} \approx 4000$ K, matching Hayashi (1961) and Henyey (1955) with $R^2 = 0.9998$.